Starter application and tests

Vasisht Gaddale | Coupon Validator | Assignment 31

The CLI runs used save-20, 199,49.5,320 and 2024-01-01 as the valid inputs. The command panels are rendered from captured terminal output.

Original application with valid input

```
$ python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: Days Since 2024: 1
```

Starter pytest collection error (excerpt)

```
$ python -m pytest -q --tb=line test_coupon_validator.py

[Stack trace frames omitted]
E     File "/workspace/scratch/d9f47fac55db/baseline/test_coupon_validator.py", line 7
E       @given(st.text() | st.none())
E       ^
E   IndentationError: expected an indented block after function definition on line 5
=========================== short test summary info ============================
ERROR test_coupon_validator.py
!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!
1 error in 0.21s
```

The supplied test functions had empty bodies, so pytest stopped during collection. The completed property tests were run on the fixed validator only. This page records a collection error and contains no before-fix Hypothesis failures.

Fixed application and tests

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Invalid inputs return None. Coupon prefixes are normalized to uppercase, cart prices must be finite and nonnegative, and expiry offsets use real calendar dates with 2024-01-01 as day 1.

Fixed application with the same valid input

```
$ python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): Prefix and Value: ('SAVE', 20)
==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): Most Expensive Item: 320.0
==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: Days Since 2024: 1
```

Fixed application with invalid input

```
$ python coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): Prefix and Value: Invalid input
==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): Most Expensive Item: Invalid input
==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: Days Since 2024: Invalid input
```

Completed Hypothesis suite

```
$ python -m pytest -q test_coupon_validator.py
.........                                                                [100%]
9 passed in 0.55s
```

Nine properties cover mixed-type inputs, valid results and invalid-input rejection across the three functions.